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SYSTEMATIC REVIEW

Terminology and Interpretation Across Neuromuscular Profiling Methods: A Systematic Review

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Abstract

Background Neuromuscular profiling (NP) provides an objective means of assessing neuromuscular function to guide targeted training interventions. However, coaching terminology used to describe NP outputs (e.g., 'explosive', 'springy') is often inconsistently defined or applied, leading to misinterpretation and limiting effective use. Variations in how NP metrics are calculated and interpreted further complicate comparisons across studies and practical applications. Despite their widespread use in sport science and performance settings, no framework currently exists to evaluate whether the terminology and interpretations associated with NP metrics accurately represent measured outputs and intended neuromuscular traits.

Objectives This review aimed to evaluate whether terms/labels used across six commonly applied NP methods—Dynamic Strength Index, Eccentric Utilisation Ratio, Force–Velocity Profile, Impulse Analysis, Reactive Strength Index, and Reactive Strength Index Modified—accurately represent the neuromuscular characteristics they are intended to measure. A secondary objective was to identify methodological limitations that may affect the interpretation, standardisation, and practical application of these methods.

Methods A systematic search of PubMed, Scopus and SPORTDiscus identified 1653 eligible studies, of which 184 met inclusion criteria. A modified quality scale was used to assess study relevance. Extracted data included reported output values, assigned terms/labels, underlying mechanisms, and supporting evidence. Results were grouped by explanatory themes, with terms/labels identified and examined based on the underlying mechanisms they were intended to represent. Findings were then synthesised to identify patterns in terminology, interpretation, and limitations.

Results A total of 209 NP methods were evaluated (147 ratio-based, 62 non-ratio-based). NP interpretation was task-dependent and influenced by methodological variability. Inconsistency was found in the terminology, definitions, and calculations used to derive NP metrics. Several studies attempted to classify athletes based on NP metrics, though few accounted for constituent components in interpretation. Commonly used terms/labels included: 'explosive/explosiveness' (indicating greater rate of force development), 'spring/springy' (suggesting effective stretch–shortening cycle utilisation), 'stiff/compliant' (representing a continuum of lower extremity stiffness), and 'strong/weak' (reflecting maximal force production capability).

Conclusions This review offers a consolidated synthesis of NP terminology, interpretation, and methodological considerations. Clarifying terminology and addressing limitations may improve communication between researchers and practitioners, supporting more standardised use of NP tools.

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Key Points

Terminology used in neuromuscular profiling (NP) methods varies considerably, leading to inconsistent interpretation of athlete capabilities across research and practice.

Methodological differences, such as variations in calculation approaches and constituent variable emphasis, substantially influence NP outcomes and complicate cross-study comparisons.

Accurate interpretation of NP results requires not only context-specific evaluation but also a standardised framework to align terminology and output variables with the neuromuscular characteristics they purport to represent.

1 Background

1.1 Perspectives on Neuromuscular Profiling

Athletes may achieve identical performance outcomes in activities like jumping and sprinting, but their movement strategy profiles can differ greatly due to anatomical, physiological, and biomechanical differences [1–3]. Such inter-individual movement variability is widely recognised by both practitioners (e.g., sport coaches, strength and conditioning specialists) [4] and researchers (e.g., sport science academics and applied scientists) [5, 6], who analyse movement variability through their unique disciplinary lenses. In practice, practitioners may place greater emphasis on observational assessment [4], and use descriptive terms/labels (e.g., 'explosive' or 'springy') to categorise movement patterns. Researchers, on the other hand, typically focus on objective evaluation of physical attributes (e.g., peak ground reaction force (GRF), rate of force development (RFD)) [5, 6] and rely on metric-based comparisons to quantify athlete profiles [5–7]. However, it is important to recognise that both groups frequently integrate both subjective (e.g., movement observations) and objective assessments (e.g., measurable outputs) when profiling athletes [4]. As Lees [4] highlights, analysing human movement presents methodological challenges, irrespective of whether subjective or objective approaches are employed. Ultimately, both practitioners and researchers aim to mechanistically explain inter-individual differences in movement strategy. This complexity highlights the need for critical evaluation and methodological rigour when interpreting profiling data, particularly within the context of neuromuscular profiling (NP). NP refers to the assessment of an individual's neuromuscular function within sport-related movements to identify mechanisms associated with movement variability, which can then inform targeted training interventions to address performance limitations [5–8].

Practitioners often rely on subjective observations and experience to group and term/label athletes based on observable temporal and kinematic outputs [4]. These outputs—such as ground contact, limb range of motion (ROM), and movement speed—are usually assessed through intuitive judgements rather than precise objective data [4]. Accordingly, practitioners group athletes into qualitative colloquial terms/labels across different sporting movements (e.g., jumping, running/sprinting, or throwing/pushing) [9–14]. To our knowledge, no formal Delphi study or systematic investigation currently exists documenting the specific terminology commonly used by practitioners to categorise athlete qualities. In the absence of such research, anecdotal sources—including blogs, books, podcasts, and social media posts [9–14]—offer observational insight into terminology trends in current coaching practice. For instance, coaches may label athletes as 'elastic/springy' or 'strength/muscle' jumpers based on the coach's perception of the movement's efficiency (i.e., does the athlete seem to be 'effortlessly bursting' through the movement or 'actively straining' through it) [9–12]. Similarly, sprint coaches may group sprinters as 'pushers' or 'pullers' depending on the athlete's body orientation, direction of force application, and perceived time on the ground [13], or term them as 'concentric', 'elastic', or 'metabolic' according to their dominant movement traits of being 'muscle-driven', 'tendon-driven', or 'substrate-driven', respectively [14]. While these subjective NP methods may offer potential insight for tailoring training interventions, their lack of empirical validation and the inconsistent use of terminology among coaches raise concerns about their reliability and application. As such, the observational insights presented here should be interpreted cautiously and are intended primarily to illustrate terminology trends in current coaching practice.

Conversely, researchers often attribute movement variability to anatomical, physiological, and biomechanical factors that influence athletes' preferred kinematic movement patterns in sport [1–3]. NP employed by researchers incorporates various objective methods that provide quantifiable measures of an athlete's neuromuscular function (e.g., peak GRF, RFD) [5–8]. These measurements are typically derived from task-specific protocols designed to isolate particular neuromuscular characteristics. Some examples are the Dynamic Strength Index (DSI), Eccentric Utilisation Ratio (EUR), Force–Velocity Profile (FVP), Impulse Analysis (IA), Reactive Strength Index (RSI), and Reactive Strength Index Modified (RSImod) [15–20]. These approaches are rooted in scientific principles and typically

employ numerical values to quantify specific aspects of the underlying mechanisms responsible for the athlete's movements. Empirically derived NP methods can be broadly categorised into ratio-based (e.g., DSI, EUR, RSI, RSImod) and non-ratio-based (e.g., FVP, IA) methods. Ratio-based NP methods yield a single value (dividing the numerator by the denominator) [15–17, 19], while non-ratio-based methods generally generate multiple numerical outputs depicting an athlete's profile [18, 20, 21]. Ratio-based methods easily allow for the establishment of numeric thresholds or cutoffs, where values can be grouped and labelled to infer the specific mechanisms or characteristics underlying movement outcomes. These groupings can then form the basis for targeted training interventions to address individual performance needs [7]. For example, DSI represents the ratio between peak force (PF) during a ballistic task (e.g., countermovement jump (CMJ)) and PF during a multi-joint isometric task (e.g., isometric mid-thigh pull (IMTP)) [22]. The ratio provides insight into the relative expression of dynamic to isometric force production [19]. DSI values are commonly grouped into ranges of ≤ 0.6 or ≥ 0.8, with corresponding terms/labels of 'low' or 'high' DSI, indicating one's magnitude of PF translation during dynamic movement [19, 23, 24]. Based on these values, targeted training interventions can be prescribed to address the specific underlying needs of athletes, such as enhancing ballistic force expression for those with 'low' DSI, or maximising force production for those with 'high' DSI [19, 23].

1.2 Terminology and Interpretation Across Neuromuscular Profiling Methods

While NP methods have been extensively researched [5–7], much of the existing literature has focused on their measurement properties, particularly reliability, across various metrics [25–30]. However, standardised frameworks for grouping and labelling NP output values remain lacking. This absence of standardisation, coupled with inconsistent terminology between practitioners and researchers, pose significant challenges in both the interpretation and application of NP results, potentially hindering effective collaboration in practice [31, 32]. For instance, Comfort et al. [24] reported that individuals with high DSI values (ranging from 0.8 to 0.92) were "notably weaker than the low DSI group". While the authors clarified that this comparison was based on relative PF during the IMTP, the phrasing may be misinterpreted as suggesting that athletes with 'high DSI' are inherently 'weak'—either in an absolute or a relative sense. This highlights the importance of providing precise contextual explanations when using classification terms such as 'weak' or 'strong' to avoid misleading interpretations. In the context of DSI, such an interpretation would be erroneous, as DSI is merely a ratio and does not, in itself, indicate an athlete's strength level. Although this example lends itself to discrete classification, it is important to recognise that the physical attribute the metric represents—in this case, strength—exists along a continuum. Therefore, it is essential to consider the constituent components of any ratio-based metric and establish consensus around their interpretation to ensure accurate application in both research and practice.

In the literature [33, 34], the term 'explosive' has occasionally been applied to athletes with high DSI values; however, such classifications are often based on logical induction rather than direct measurement of explosive-related variables (e.g., RFD, peak power (PP)). While researchers generally interpret 'high DSI' as a ratio describing the balance between dynamic and isometric force capabilities, it does not inherently reflect explosive performance. Nonetheless, practitioners may interpret this label at face value, assuming that a 'high DSI' equates to 'explosive' qualities. This interpretation is misleading since DSI does not account for temporal aspects of force production, making its use as a proxy for 'explosiveness' inappropriate. Such discrepancies underscore the need for a standardised approach to NP that accurately conveys the intended meaning of terms/labels, ensuring that both practitioners and researchers can interpret and apply findings appropriately, and ultimately facilitating more effective collaboration [31, 32].

The EUR purportedly evaluates stretch–shortening cycle (SSC) efficiency [17], typically assessed through jump height (JH) [35], though some studies have instead utilised PP [17, 36]—an instantaneous variable that does not directly reflect vertical displacement and may therefore misrepresent the construct the EUR was originally intended to assess. A greater EUR is generally suggested to indicate 'superior SSC function', as shown in studies where participants exhibited enhanced performance in CMJ compared to squat jump (SJ) [37–42]. Contrastingly, others reported that lower EUR values represented 'better SSC function' [43, 44], primarily due to different calculation methods—such as the ratio of maximal power output in SJ to CMJ [43] and net impulse [44]—as opposed to the more common method of calculating EUR as CMJ height divided by SJ height [35]. Such inconsistencies highlight the need for a deeper examination of each metric's calculation methods, the underlying mechanisms behind metric values, and the interpretation of their ranges. Moreover, EUR values above 1.0 have been labelled as 'power-dominant', while values below 1.0 are seen as 'strength-dominant' [45]. However, the PF obtained merely indicates instantaneous force production during the entire take-off phase and may not accurately reflect the terms 'power' or 'strength', as used to define individuals with the respective EUR values. The terminology may have been incorrectly employed, possibly due to a misunderstanding of the underlying principles of what the metric values truly represent. Interestingly, an increase in EUR values was

suggested to represent a shift from 'strength-dominant' to 'power-dominant' movement patterns [45]. Noteworthily, the interpretation of a property shift given a subtle change in value (i.e., difference of only 0.14) may raise questions, especially considering the absence of reliability assessments to ascertain if the change was meaningful [46, 47]. While the authors described a shift in 'movement patterns', it is important to note that EUR is derived from discrete outcome variables (e.g., CMJ and SJ height), and does not capture the temporal aspects of movement. Without supporting temporal analysis (e.g., force–velocity characteristics), such interpretations lack sufficient evidence and should be considered speculative. Overall, assigning terms/labels based on set cut-off values can oversimplify the complexity of human movement and performance. A more nuanced approach, considering the constituent components of the metric [48] and ensuring robust reliability measure [47], will enable practitioners to develop a comprehensive understanding of athletic capabilities and needs.

2 Objectives of the Review

The use of terms like 'explosive', 'springy', or 'stiff' to represent certain ranges of NP output values can be misleading—especially when such terms/labels are interpreted without a clear understanding of what the underlying metrics truly represent. Moreover, the physical qualities these terms aim to describe—such as force production, velocity, and stiffness—exist along a continuum, rather than as discrete categories, making rigid classification potentially misleading in both research and applied settings. To date, no empirical studies have directly examined the alignment between practitioner-derived labels and measured NP characteristics. This disconnect highlights the need for clearer frameworks to support consistent interpretation and application of NP results in training and performance settings. Compounding this issue, variations in how NP metrics are calculated can further contribute to inconsistent interpretations. These discrepancies are problematic because they suggest that similar NP values may be interpreted differently depending on the calculation method, which can affect how athletes are profiled, trained, and monitored. As outlined in Sect. 1.2, such inconsistencies highlight the need for greater clarity and standardisation in how NP metrics are interpreted and applied.

This systematic review addresses a currently unexamined gap in the literature by evaluating whether the terminology and interpretations used in NP methods accurately represent the measured outputs and intended neuromuscular traits. It also identifies key methodological limitations to support clearer interpretation and more consistent application of these profiling tools. This involves examining the:

1. Range of output values across NP methods.
2. Terms/labels assigned to each group of values.
3. Underlying mechanisms proposed to differentiate between these groups.
4. Justification and evidence supporting the assigned terms/labels and mechanisms, whether measured quantitatively or inferred qualitatively.

This review seeks to enhance consistency and communication regarding NP, which can in turn narrow the gap between applied and research domains. For practitioners, findings will support more informed decision-making in practical training interventions and improve the calibration of profiling methods towards a more objective perspective. For researchers, this review will enhance future study designs and identify additional applied research questions. Ultimately, such harmonisation is essential for developing effective and targeted training strategies that can be confidently implemented to optimise athletic performance.

3 Methods

3.1 Study Design and Literature Search

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [49]. The review protocol was not pre-registered. Prior to initiating the literature search, a scoping phase was conducted to identify the NP methods of interest. Based on this process, six NP methods were selected: DSI, EUR, RSI, RSImod, FVP, and IA. These methods were agreed upon by the research team due to their established use in sport science applications and their relevance to sporting movements (e.g., jumping, sprinting) [5–8]. Search terms were formulated accordingly and agreed upon by investigators KC, PC, and SL. A comprehensive search was conducted in PubMed, Scopus, and SPORTDiscus in February 2023, with an update in September 2024. The search strategy is presented in Table 1.

The main investigator (KC) conducted the online search and uploaded the results into the Covidence systematic review software (Veritas Health Innovation, Victoria, Australia) (refer to Fig. 1 for PRISMA flow chart). Duplicate articles were removed, and two independent reviewers (KC and DL) screened the remaining titles and abstracts for relevance. Studies involving any NP methods with quantifiable output were included. A full-text review was then performed by KC and JF to further determine eligibility. Studies that met the predefined inclusion criteria were included.

Table 1 Search strategy syntaxes and results

PubMed	
"Dynamic strength deficit"[TIAB] OR "dynamic strength index"[TIAB] OR "eccentric utili*"[TIAB] OR "explosive strength deficit"[TIAB] OR "force velocity"[TIAB] OR "Isometric midthigh pull"[TIAB] OR "isometric mid thigh pull"[TIAB] OR "load velocity"[TIAB] OR "optimal drop height"[TIAB] OR "pre stretch augmentation"[TIAB] OR "prestretch augmentation"[TIAB] OR "reactive strength"[TIAB] OR "stretch shortening cycle"[TIAB] NOT (meta-analysis[Filter] OR review[Filter] OR systematicreview[Filter])	3497 results (filtered for English) *Last search date: 10 Sept 2024*
SPORTDiscus	
TI ("dynamic strength deficit" OR "dynamic strength index" OR "eccentric utili*" OR "explosive strength deficit" OR "force velocity" OR "isometric midthigh pull" OR "isometric mid thigh pull" OR "load velocity" OR "optimal drop height" OR "pre stretch augmentation" OR "prestretch augmentation" OR "reactive strength" OR "stretch shortening cycle") OR AB ("dynamic strength deficit" OR "dynamic strength index" OR "eccentric utili*" OR "explosive strength deficit" OR "force velocity" OR "isometric midthigh pull" OR "isometric mid thigh pull" OR "load velocity" OR "optimal drop height" OR "pre stretch augmentation" OR "prestretch augmentation" OR "reactive strength" OR "stretch shortening cycle")	1872 results (filtered for English, Academic Journal, Peer Reviewed) *Last search date: 10 Sept 2024*
Scopus	
TITLE-ABS ("dynamic strength deficit" OR "dynamic strength index" OR "eccentric utili*" OR "explosive strength deficit" OR "force velocity" OR "isometric midthigh pull" OR "isometric mid thigh pull" OR "load velocity" OR "optimal drop height" OR "pre stretch augmentation" OR "prestretch augmentation" OR "reactive strength" OR "stretch shortening cycle") AND (LIMIT-TO (SRCTYPE, "j")) AND (LIMIT-TO (LANGUAGE, "English")) AND (EXCLUDE (DOCTYPE, "re"))	5205 results (filtered for English, Academic Journal, Excluding Reviews) *Last search date: 10 Sept 2024*

References of included studies were also screened to identify further relevant studies. Discrepancies were discussed and resolved by both reviewers (KC and JF). If a consensus could not be achieved, discrepancies were resolved with the aid of a third reviewer (SL). Eligible full-text studies underwent data extraction and quality assessment by KC and SL, with consensus reached through discussion.

3.2 Inclusion and Exclusion Criteria

The inclusion criteria were: (1) original English full-text peer-reviewed articles involving healthy human participants with no restrictions to age or sex; (2) evaluated NP methods involving jumping, running/sprinting, or throwing/pushing motions; (3) provided terms/labels of the NP metric(s) and/or provided justifications as to how the outcome metric(s) were interpreted/categorised. Articles excluded were: (1) non-English materials such as abstracts, books and/or conference proceedings, and outside the scope of human involvement; (2) tests administered did not involve jumping, running/sprinting, or throwing/pushing motions; (3) interpretation/categorisation could not be elucidated from the study; (4) assessed under manipulated environment or altered physical conditions. Table 2 provides the detailed inclusion and exclusion criteria for study selection.

3.3 Assessment of Study Quality

Conventionally, when evaluating the methodological quality of included studies, assessment tools such as the Downs and Black scale, Physiotherapy Evidence Database (PEDro), or the Cochrane Collaboration tool are commonly adopted across healthcare research (i.e., dietetics, physiotherapy) [50]. Owing to the distinctive novelty of this systematic review, whereby the emphasis is on content rather than methodology, existing quality assurance tools may be inappropriate for non-healthcare research as such (i.e., sport science), because they tend to be scored low on these scales [51]. Based on proposed guidelines [50], a customised template for quality assessment was constructed by the authors (KC, SL, and PC) to suit the needs of this systematic review (Annex A). This customised tool objectively assesses the selected studies for both content and methodological quality. It comprises ten article evaluation items, seven of which are content-related and three others being methodology-related, with 1 point scored for each 'yes' and 0 for each 'no' indicated. The minimum and maximum score each article can achieve ranges from 0 to 10, with higher scores indicating greater content and methodological quality. Scores <4 are categorised as 'poor', ≥ 4 to <6 as 'fair', ≥ 6 to <9 as 'good', and ≥ 9 to 10 as 'excellent'.

3.4 Data Extraction and Analysis

Data extraction involved a systematic approach to gather pertinent information from eligible studies, aiming to examine the (1) range of output values across NP methods, (2) terms/labels assigned to each group of values, (3) underlying mechanisms proposed to differentiate these groups, and (4) justification and evidence supporting the assigned terms/labels and mechanisms, whether measured quantitatively or

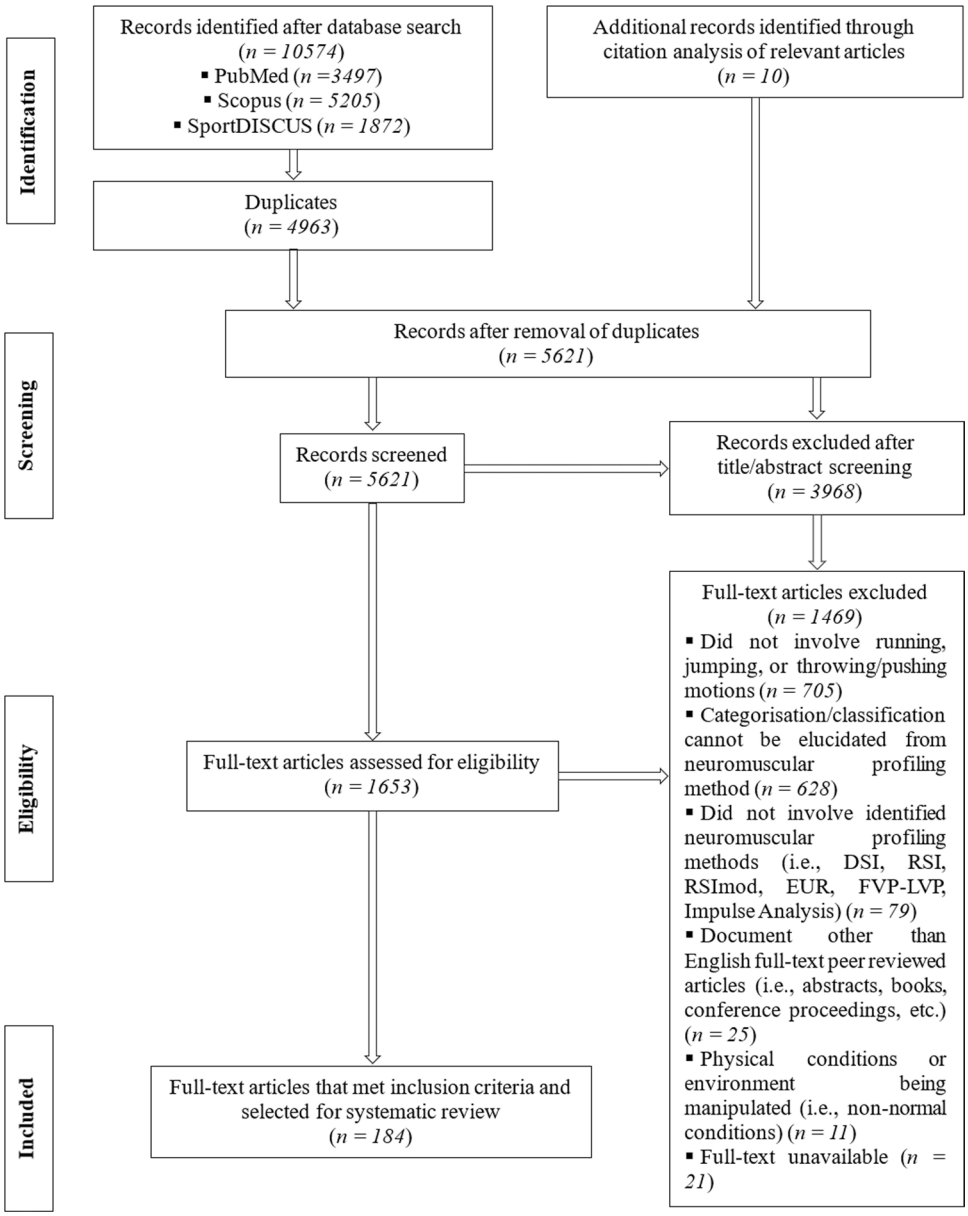

Fig. 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow chart

inferred qualitatively. The following data were extracted from each eligible study via the Covidence software:

i. *General information:* Title, lead author, year of publication.
ii. *Methodological elements:* Study aims, design, experimental details (if applicable), population profile (i.e., total participants, sex, age, height, body mass, description).
iii. *NP method(s):* Name, definition and/or measurement methods, reported output values, derived terms/labels and mechanisms, origin of terms/labels and mechanisms (quantitative vs. qualitative), training interventions.
iv. *Reliability and measurement error:* Metrics and findings related to the reliability of the NP methods and their associated measurement errors.

The extracted data were then systematically assessed, categorised, and analysed to qualitatively address the research question. Four explanatory themes were identified based on the terms/labels and mechanisms derived from the findings:

1. *Adaptation*—encompassed mechanisms associated with both neuromuscular and morphological changes. This included the efficiency/inefficiency of SSC and increased/reduced neuromuscular adaptations (i.e., motor unit recruitment, firing frequency, synchronisa-

Table 2 Study inclusion–exclusion criteria

Inclusion criteria	Reasons for exclusion
Article type and language	Non-English full-text studies such as abstracts, book chapters, conference proceedings, errata, letters to the editor, meta-analyses, opinion pieces, reviews, surveys, systematic reviews, narratives of testing battery, technical notes, training or rehabilitation programs, etc.
Studies published from inception to 10 September 2024 were considered for inclusion, including studies published in the Epub ahead of print format	Publication date after 10 September 2024 or not yet available as Epub ahead of print
Population involved healthy human participants	Outside the scope of human involvement including the following: – Animals and/or insects – Computer simulations and/or modelling – In-vitro (conducted outside living organism) – Non-voluntary muscular contraction (e.g., electrically stimulated muscle) – Systems other than the musculoskeletal system (e.g., smooth muscle, cardiac muscle, respiratory muscle, retina, masticatory system)
Outcome measures involved one or more *pre-identified neuromuscular profiling methods that provided some form of classification or categorisation **Pre-identified—Refers to NP methods selected prior to the literature search based on an investigator-led scoping phase. Six methods (DSI, EUR, RSI, RSImod, FVP, IA) were chosen for their established use in sport science and relevance to common sporting movements (e.g., jumping, sprinting)*	Did not assess and report any pre-identified neuromuscular profiling methods such as: – Dynamic Strength Index (DSI) – Dynamic Strength Deficit (DSD) – Explosive Strength Deficit (ESD) – Reactive Strength Index (RSI) – Reactive Strength Index Modified (RSImod) – Eccentric utilisation ratio (EUR) – Pre-stretch augmentation percentage – Reactive strength – Stretch shortening cycle performance – Force velocity profiling or testing – Examining force–velocity relationships or characteristics – Load velocity profiling or testing – Examining load–velocity relationships or characteristics Classification or categorisation such as the following was not reported: – Numerical values—DSI, RSI, RSImod, EUR – Categories or descriptives—force/velocity deficiency; high/moderate/low; spring/power – Movement specific descriptions—shallow/deep, fast/slow
Profiling methods involved running, jumping and/or throwing pushing motions	Profiling methods administered did not involve running, jumping or throwing/pushing motions
Physical conditions or environment were not altered	Physical conditions or environment being manipulated (e.g., altitude intervention, blood-flow restriction, non-normal conditions)
Classification can be elucidated from the study through the article	Did not provide distinct categorisation of the neuromuscular profiling metric(s) (e.g., 'explosive', 'stiff', 'springy', 'force-dominant' vs. 'velocity-dominant', 'strong' vs. 'weak'), and/or did not provide justifications as to how the outcome metric(s) were interpreted and/or implicitly categorised (i.e., explained the mechanisms behind neuromuscular profiling metric outcome)

tion, RFD, muscle cross-sectional area, fascicle length/thickness, tendinous adaptation).

2. *Movement Strategy*—describing increased/reduced jump performance or take-off velocity, faster/slower or deeper/shallower countermovement, exhibiting bimodal/unimodal profile, and joint-strategies (i.e., hip-, knee-, or ankle-dominant).
3. *Stiffness*—providing information on whether the lower extremity was compliant/stiff.
4. *Strength*—demonstrating enhanced/reduced force production or attenuation (i.e., ability to tolerate eccentric loading) and separating stronger/weaker individuals.

Terms/labels and mechanisms were identified either through quantitative measures (e.g., recorded isometric force production, stiffness measurements), or qualitative inductions supported by logical reasoning (e.g., improved SSC abilities due to maturation, increased stiffness due to training adaptations). Justifications and evidence were further extracted and systematically tabulated.

Additionally, reported NP method values were extracted to denote the groups (i.e., output values) corresponding to the identified terms/labels and mechanisms in each study, along with the measurement methods used.

4 Results

4.1 Search Results

Out of the 10574 studies identified across the databases, 4963 duplicates were removed and 3968 articles were excluded after title and abstract screening, leaving 1643 studies. An additional ten studies were identified through citation analysis, resulting in 1653 studies assessed for eligibility. A further 1469 studies were excluded for not meeting the inclusion criteria. Overall, 184 studies were included and evaluated (Fig. 1), out of which 209 NP methods were recorded (i.e., one paper could investigate more than one profiling method). Six NP methods were identified based on their concepts and/or measurement methods across all evaluated studies and were further organised as ratio-based NP methods (DSI *(n = 12)*; EUR *(n = 19)*; RSI *(n = 72)*; RSImod *(n = 44)*) and non-ratio-based NP methods (FVP *(n = 42)*; IA *(n = 20)*). Supplementary Tables 1.1–1.6 (Electronic Supplementary Material (ESM)) provide a detailed summary of findings across the six NP methods, including study population, definition and measurement method, reported values, reliability results, and derived themes, terms/labels, and mechanisms identified from the included studies, along with possible training interventions.

4.2 Quality Assessment

A total of 184 studies were assessed for content and methodological quality using the modified quality assessment tool (Annex A) by two reviewers (KC and SL). Discrepancies were resolved through discussion between the reviewers allowing consensus to be reached. If a paper investigated more than one NP method, quality assessment was evaluated focusing on the profiling method that was primarily discussed within the study. The included studies had an average ± standard deviation (SD) score of 6.1 ± 1.5 (range 2–9), indicating overall 'good' content and methodological quality (Table 3). The detailed quality assurance assessment scores are presented in Supplementary Tables 2.1–2.6 (ESM).

4.3 Best Evidence Synthesis

Supplementary Tables 3.1–3.6 (ESM) present the extracted themes, terms/labels, mechanisms, output values, and their respective measurements methods, along with supporting evidence from each study. Evidence was categorised based on whether findings were directly measured (quantitative) or inferred (qualitative). Key results are reported as follows.

4.3.1 Dynamic Strength Index

Overall, 12 studies assessed DSI and obtained a quality score of 7.0 ± 1.2 (range: 5–9), indicating 'good' content and methodological quality (Table 3; Supplementary Table 1.1 (ESM)). Out of these studies, ten reported absolute DSI values [23, 24, 33, 34, 48, 52–56], one did not report the constituent variables (i.e., CMJ PF and IMTP PF) resulting in DSI [6], and four presented reliability statistics [19, 23, 24, 52]. Reported DSI values ranged from 0.46 to 0.92 (Supplementary Table 3.1 (ESM)). Themes identified were primarily related to *'strength'*, with associated labels such as 'strong', 'weak', 'force-dominant', and 'explosive'. Higher DSI values (0.65–0.91) were associated with mechanisms such as efficient SSC utilisation [53, 54] and greater lower extremity stiffness [53], and labelled as 'explosive' [33, 34, 55]. Lower DSI values (0.46–0.76) were associated with greater strength qualities [6, 19, 23, 34, 48, 52], and labelled as 'strong' [24, 33, 54] or 'force-dominant' [56], while higher DSI values (0.76–0.92) were associated with

Table 3 Overall quality assurance scores obtained by the respective neuromuscular profiling methods

Neuromuscular profiling method	Average score	Standard deviation	Lowest score	Highest score
Dynamic Strength Index ($n = 12$)	7.0	1.2	5	9
Eccentric Utilisation Ratio ($n = 19$)	5.5	1.7	2	8
Force–Velocity Profile ($n = 42$)	7.0	1.1	4	9
Impulse Analysis ($n = 20$)	6.2	1.8	3	9
Reactive Strength Index ($n = 72$)	5.4	1.3	2	7
Reactive Strength Index Modified ($n = 44$)	6.4	1.2	4	9
Overall ($n = 209$)	6.1	1.5	2	9

Score Interpretation—Poor: < 4; Fair: ≥ 4 to < 6; Good: ≥ 6 to < 9; Excellent: ≥ 9 to 10

mechanisms such as reduced force production [19, 23, 52] and were labelled as 'weak' [24, 54].

4.3.2 Eccentric Utilisation Ratio

EUR was measured across 19 studies, with a quality score of 5.5 ± 1.7 (range: 2–8), demonstrating 'fair' assessment quality (Table 3; Supplementary Table 1.2 (ESM)). All [17, 33, 36–45, 57, 58] except five studies reported numerical EUR values, eight did not report the constituent components (i.e., SJ and CMJ height/PP/PF) resulting in EUR [17, 35, 44, 58–62], and one presented reliability statistics [17]. Due to varied methods of calculation (i.e., expressed as a ratio, percentage, or absolute difference), a common range of EUR values could not be established. Therefore, values were synthesised according to magnitude (greater or lower) (Supplementary Table 3.2 (ESM)). Identified themes centred around *'adaptation'*—reflecting improvements commonly attributed to efficient SSC function—*'stiffness'*, and *'movement strategy'*. Greater EUR values were commonly associated with more efficient SSC function [35, 37–42, 45, 58–62], and one study used the label 'springy' [59], while lower EUR values were linked to reduced SSC efficiency [37–42]. In contrast, two studies reported opposing findings, associating lower EUR values with more efficient SSC function and greater EUR values with less efficient SSC function [43, 44]. Mechanisms related to stiffness were also described: greater EUR values were associated with lower extremity stiffness and labelled as 'compliant' [37, 39–41], while lower EUR values were associated with greater stiffness and labelled as 'stiff' [33, 37, 39–41]. Greater EUR values were further linked to movement strategies described as 'hip- and/or ankle-dominant' [57, 59], and associated with enhanced force production [37, 42, 45], while lower EUR values were associated with the label 'strength-dominant' [45]. One study described greater EUR values as indicative of being 'less explosive' [36].

4.3.3 Reactive Strength Index

RSI was assessed in 72 studies achieving a quality score of 5.4 ± 1.3 (range 2–7), representing 'fair' quality (Table 3; Supplementary Table 1.5 (ESM)). Of these, 65 reported numerical RSI values [40, 63–126], 33 did not report both constituent variables that make up RSI (i.e., JH and ground contact time (GCT)) [6, 39, 64, 69–72, 76–78, 82, 85, 88, 89, 92–94, 96–98, 100, 104, 113, 114, 117, 119, 121, 127–132], and 15 presented reliability statistics [65, 66, 70, 72, 77, 88, 89, 101, 105, 111, 116, 122, 124, 126, 131]. A common range of RSI values could not be established due to the varying measurement methods used across studies (i.e., drop jump (DJ), rebound jump, flight time). Therefore, RSI values were evaluated through their magnitude (greater or lower) (Supplementary Table 3.3 (ESM)). Identified themes included *'adaptation'*, *'movement strategy'*, *'stiffness'*, and *'strength'*. Greater RSI values were commonly reported following intervention and were attributed by study authors to a range of neuromuscular adaptations (e.g., increased motor unit recruitment, firing frequency, and synchronisation) [64, 69, 72, 73, 76, 79, 82, 83, 103, 113, 117, 121, 127]. These adaptations were described as contributing to greater RFD [69, 107], and often linked to labels such as 'explosiveness' [131] or 'explosive abilities' [100, 125, 126]. Greater RSI values were also associated with mechanisms such as efficient SSC utilisation [40, 71–75, 79, 80, 83, 84, 88, 89, 92–94, 97, 99, 106, 108, 109, 111, 114, 120, 122, 124, 128–130], with some studies describing this as 'good SSC function' using the label 'spring-like' [106, 109, 128]. Conversely, lower RSI values were associated with mechanisms such as inefficient SSC utilisation or 'poor SSC function' [75, 79, 80, 92, 93, 97, 106, 109–111, 116]. Several studies described greater RSI values in relation to movement strategies such as increased JH [70, 118, 119, 124], reduced GCT [74, 85, 87, 99, 103, 107, 120], a combination of both [68, 108], reduced ROM [85, 102], or stiff landing strategies [85, 120]. Greater RSI values were commonly associated with increased lower extremity stiffness and described using the label 'stiff' [40, 69, 72–75, 77, 79, 80, 82–84, 87, 88, 90, 92–94, 96, 102, 104, 108, 113, 114, 119, 123, 128–130], whereas lower RSI values were associated with reduced limb stiffness or described as 'compliant' [40, 84, 92, 93, 102, 104]. Lastly, greater RSI values were linked to enhanced force production or attenuation [6, 63, 65, 67, 68, 71, 76–80, 86, 90, 95–98, 112, 113, 115, 117–119, 124, 127, 132], and described as 'strong' [63, 67, 68, 76, 78–80, 86, 95, 97, 112, 113, 118, 119], while lower RSI values were associated with reduced force production or attenuation [63, 67, 68, 75, 78–81, 86, 96, 98, 110, 115], and labelled as 'weak' [63, 67, 68, 78–81, 86, 110].

4.3.4 Reactive Strength Index Modified

Forty-four studies examined RSImod, averaging 6.4 ± 1.2 (range 4–9), indicating 'good' content and methodological quality (Table 3; Supplementary Table 1.6 (ESM)). The majority of the studies (41 out of 44) provided numerical RSImod values [15, 23, 26, 44, 48, 101, 105, 111, 124, 126, 133–163], 12 did not report both constituent components of RSImod (i.e., JH and time to take-off (TTT)) [35, 44, 48, 136, 151–153, 161–165], and 22 studies reported reliability statistics [15, 23, 26, 48, 101, 105, 111, 124, 126, 134, 137, 138, 141, 145–149, 152, 155, 160, 162]. Because of the varying measurement methods and calculations used for RSImod (i.e., JH, jump distance, flight time), a common range of values could not be determined. RSImod values were hence evaluated through their magnitude (greater or

lower) (Supplementary Table 3.4 (ESM)). As interpreted across the included studies, RSImod was associated with the recurring themes of *'adaptation'*, *'movement strategy'*, and *'strength'*, reflecting mechanisms such as efficient SSC function, greater RFD, rapid unloading and enhanced force production. Greater RSImod values were associated with efficient SSC utilisation [35, 44, 124, 133, 141, 145, 149, 154] and increased RFD [15, 134, 140, 143, 144, 153, 154, 162, 165], and thus were labelled as 'explosive' [15, 126, 134, 143, 162, 163, 165]. In contrast, lower RSImod values were associated with inefficient SSC utilisation [44, 149, 154] and reduced RFD production [154]. Greater RSImod values were also linked to quicker countermovement (i.e., faster unloading) [23, 26, 111, 133, 137, 139, 140, 142–144, 155, 157, 158, 161, 165], and increased take-off velocity, contributing to improved jump performance [26, 111, 124, 139, 142, 147, 149–153, 155, 160, 161]. Conversely, lower RSImod values were associated with slower unloading [111, 137, 139, 142, 165] and reduced jump performance [23, 111, 139, 142, 147, 149, 153]. There was mixed evidence regarding countermovement depth: some studies indicated that deeper countermovement (i.e., greater ROM) was associated with higher RSImod values [26, 145, 148], while others reported that shallower countermovement (i.e., lesser ROM) produced greater RSImod [150, 153, 155, 156]. Similarly, greater RSImod values were described as 'stiff' in some studies [134, 141, 159], while others suggested otherwise (i.e., lower RSImod as 'stiff') [148]. Furthermore, greater RSImod values were associated with enhanced force production capabilities [23, 48, 134–136, 138, 139, 141, 145–149, 153–155, 158–162], and were labelled as 'strong' [48, 134, 138, 141], while lower RSImod values were associated with reduced force production capabilities [134–136, 138, 154] and labelled as 'weak' [134, 138].

4.3.5 Force–Velocity Profile

FVP was investigated in 42 studies, scoring 7.0 ± 1.1 (range 4–9), indicating 'good' quality (Table 3; Supplementary Table 1.3 (ESM)). Most studies (40 out of 42) reported FVP variables [18, 21, 56, 166–202], and only four shared reliability statistics [171, 172, 190, 195]. Across the included studies, themes of *'movement strategy'* and *'strength'* were most prominent. Profiles with steeper force–velocity slope (*FV*slope) (i.e., more negative values) and force–velocity imbalance (*FV*imb) of more than 100% of the optimal FVP were considered 'force-oriented', 'force-dominant', or exhibiting a 'velocity-deficit', reflecting greater force capability [18, 21, 168, 169, 173–176, 178, 180–190, 192, 195–198, 200, 202–204] (Supplementary Table 3.5 (ESM)). In contrast, profiles with gentler *FV*slope (i.e., less negative values) and *FV*imb of less than 100% of the optimal FVP were described as 'velocity-oriented', 'velocity-dominant', or exhibiting a 'force-deficit', reflecting greater velocity capability [18, 21, 56, 166, 170, 174–178, 181, 183, 185, 186, 193–196, 198, 200–202]. Some sources also used the terms 'terrestrial' to describe 'force-oriented' profiles, and 'aerial' to describe 'velocity-oriented' profiles [199], with the latter occasionally associated with labels such as 'explosive' [179, 189].

4.3.6 Impulse Analysis

A total of 20 studies analysed force–time (*F*–*T*) curve variables across CMJ (15 studies) [20, 23, 44, 143, 147, 154, 155, 163, 164, 205–210] and DJ (five studies) [106, 108–110, 211], and recorded a score of 6.2 ± 1.8 (range 3–9), displaying 'good' quality (Table 3; Supplementary Table 1.4 (ESM)). Given that the IA methods examine patterns and relationships across diverse temporal and mechanical dimensions, findings spanned a broad spectrum of themes. All studies described results obtained from their respective analysis [20, 23, 44, 106, 108–110, 143, 147, 154, 155, 163, 164, 205–211], while only one reported reliability results [210]. Greater jump performance (e.g., greater JH or take-off velocity), as well as higher RSI and RSImod values, were often associated with efficient SSC utilisation [20, 44, 106, 108–110, 154, 205, 208, 211], and in some cases described using the label 'spring-like' [106, 109, 110] (Supplementary Table 3.6 (ESM)). These profiles also demonstrated greater RFD [20, 143, 154] and were described using the label 'explosive' [143, 163]. Profiles with enhanced force production (i.e., 'stronger' profiles) often employed quicker countermovement to produce greater forces earlier in the jump phases, contributing to improved performance outcomes [143, 205–207]. However, due to conflicting evidence across studies [20, 23, 44, 110, 154, 155, 164, 205–207, 209–211], no consistent pattern could be established regarding the influence of countermovement depth (i.e., increased or reduced ROM), or the potential role of 'bimodal' versus 'unimodal' profiles in determining an optimal strategy.

5 Discussion

NP provides an objective means of assessing an individual's neuromuscular function within sport-related movements to identify mechanisms associated with movement variability, which can then inform targeted training interventions to address performance limitations [5–8]. However, the terms/labels used to describe NP metrics are often inconsistently defined or misinterpreted by both practitioners and researchers, which can lead to inaccurate profiling and misapplications in training or research contexts. In addition, variations in how NP metrics are calculated can produce discrepancies in reported values, further complicating interpretation and

comparison. These issues are compounded by the fact that physical attributes—such as strength and stiffness—exist along a continuum, making rigid classifications potentially misleading. To address these concerns, this systematic review evaluated whether the terminology and interpretations used across NP methodologies accurately reflect the measured outputs and intended neuromuscular traits, while also identifying key methodological limitations that may hinder consistent and meaningful application. Overall, each method was conceptually framed to assess specific neuromuscular characteristics or task demands. However, the findings revealed considerable overlap in interpretive themes across methods, and little consensus regarding established output value ranges, likely due to variation in methodology, interpretation, and population characteristics (e.g., age, strength training history), which can substantially influence NP outcomes.

5.1 Terminology and Interpretation Across Neuromuscular Profiling Methods

5.1.1 Ratio-Based Neuromuscular Profiling Methods

5.1.1.1 Dynamic Strength Index The DSI provides an insight into the relative expression of dynamic to isometric force production, and is typically represented as the ratio of PF produced during a ballistic task (e.g., CMJ) relative to that produced during an isometric task (e.g., IMTP). Terms/labels and mechanisms related to the theme of *'strength'* (i.e., 'strong' or 'weak') were consistent. This is not surprising given that DSI is a ratio-based formula, where its constituent components (i.e., force capacities quantified dynamically and isometrically) can provide insight into an individual's strength profile. However, it is crucial to understand that DSI itself does not determine strength levels. Instead, strength levels were quantified using PF obtained during multi-joint isometric strength testing, as shown across studies [19, 23, 24, 33, 34, 54]. Rather than measuring absolute or relative strength, the DSI ratio offers insight into an athlete's neuromuscular adaptation, highlighting their bias towards maximal or ballistic force production. Coaches can then interpret DSI within the context of training phases, adjusting programming to maintain or shift the athlete's neuromuscular profile as needed.

Based on the findings (Supplementary Table 3.1 ESM), individuals with lower DSI (0.46–0.76) possessed greater strength qualities [6, 19, 23, 34, 48, 52], and were regarded as 'strong' [24, 33, 54] or 'force-dominant' [56]. Those with higher DSI (0.76–0.92) exhibited reduced strength qualities [19, 23, 52] and were considered 'weak' [24, 54]. It is important to note that the classification as 'strong' or 'weak' was not determined by the absolute value of DSI itself, but rather by the underlying force production capabilities of the individuals [6, 19, 23, 24, 33, 34, 48, 52, 54, 56]. However, a low DSI value does not always mean that an individual is strong—on an absolute or relative level—nor does a high DSI value necessarily indicate that an individual is weak [24, 47]. Strength classifications, such as 'strong' or 'weak', were typically based on PF measured during the IMTP [19, 23, 24, 33, 34, 54], not the DSI itself. Therefore, labelling individuals with 'high DSI' as 'weak', or vice versa, is misleading, as the DSI only represents the proportion of maximal isometric force utilised during a ballistic task, without providing a direct measure of absolute or relative strength. This underscores the need to carefully consider the constituent components (i.e., dynamic- and isometric-task PF) when contextualising DSI values [48]. Particular emphasis should be placed on isometric PF, as isometric tests are strong predictors of maximal strength [212, 213] and offer more precise insight into one's strength profile. Across the included sources, all except one reported the constituent components of DSI [6, 23, 24, 33, 34, 48, 52–56] (Supplementary Table 1.1, ESM), allowing better perspective and interpretation of the resulting values. DSI may offer insight into strength qualities, but its interpretation requires careful consideration of the underlying dynamic and isometric PF. Without this context, DSI alone may lead to inaccurate conclusions about an athlete's strength.

The DSI thresholds observed in this review (low: ≤ 0.76; high ≥ 0.76) differed slightly from those commonly reported in the literature (low: ≤ 0.60; high: ≥ 0.80) [19]. This discrepancy likely stems from methodological variations in how DSI is assessed, particularly the selection of dynamic and isometric tasks, as well as the inclusion of both upper and lower body assessments. For instance, the thresholds originally proposed were primarily derived from lower body assessments using CMJ PF/IMTP PF [23, 24, 33, 48, 53–55]. In contrast, this review included alternative combinations such as SJ PF/IMTP PF [19], and CMJ PF/Isometric Squat (ISqt) PF [6], as well as upper body measures like the ballistic bench press PF/isometric bench press PF [34]. These differences in test modality and the specific exercises selected can lead to variations in DSI values, highlighting the importance of interpreting thresholds relative to the specific assessment protocol.

Values ranging from 0.65 to 0.91 were said to demonstrate efficient SSC utilisation [53, 54] and were inferred as 'explosive' in some studies [33, 34, 55] (Supplementary Table 3.1, ESM). The term 'explosive' was derived implicitly through logical induction without measurement of any 'explosive' related variables by the authors [33, 34, 55], and should hence be interpreted with caution. Though taking the constituent components of DSI into account, it would nevertheless seem logical for individuals exhibiting greater DSI values (i.e., greater dynamic to isometric PF ratio) to gravitate towards having enhanced ability to apply force in

dynamic tasks (i.e., more efficient SSC utilisation). However, the use of PF in the DSI formula does not reflect one's ability to express force across time (i.e., impulse = Δforce × Δtime), which from an athletic performance standpoint, is a better predictor of dynamic performance compared to PF [52]. Besides, no correlation was observed between DSI and early IMTP force production (which represents rapid force generation) [214], further suggesting that DSI is not linked to the term 'explosive'. Therefore, the use of the term 'explosive' may not be warranted, especially since DSI provides no temporal aspects of force production.

In addition, individuals with greater DSI values (≥ 0.75) were said to exhibit greater lower extremity stiffness, and hence termed 'stiff' [53]. Stiffness is the body's ability to resist deformation in response to force, allowing it to store and return elastic energy effectively [215]. It is important to note that stiffness was estimated using CMJ [53], which serves only as a surrogate measure of stiffness. Unlike tasks involving an initial impact phase and rapid SSC (e.g., repeated hops, DJ), methods adopting CMJ and similar jumps do not accurately capture how the leg-spring mechanism is loaded during sporting activities [216]. Although a statistical relationship between stiffness and DSI has been established [53], classifying individuals with high DSI as 'stiff' would be misleading, as the components of DSI do not provide specific information about stiffness.

In summary, while the DSI ratio offers valuable insight into an athlete's balance between maximal isometric and dynamic force capacities, it should not be misinterpreted as a direct measure of absolute or relative strength. Instead, it serves as an indicator to guide training focus towards either maximal and/or ballistic force production depending on the stipulated adaptation goals of the training intervention. Terms/labels such as 'strong', 'weak', 'explosive' or 'stiff' should be avoided, as they can be misleading given that constituent components of the DSI do not provide information regarding these variables. DSI alone does not offer a comprehensive athletic profile, as it overlooks the temporal aspects of force production and focuses only on instantaneous variables. This highlights the need to complement DSI with additional NP methods or performance data to provide a more holistic view of an athlete's capabilities.

5.1.1.2 Eccentric Utilisation Ratio The EUR is intended to assess how effectively an individual's countermovement (i.e., downward movement) contributes to the performance of an SSC task. EUR is often used as an indirect indicator of SSC function and is presented as a ratio, percentage, or difference across variables obtained from an SSC task to that without an SSC (e.g., CMJ height:SJ height; [CMJ height/SJ height]/SJ height × 100%; CMJ height − SJ height). Based on the findings, EUR was associated with the theme of *'adaptation'*, reflecting improvements commonly attributed to efficient SSC function across the included studies.

Most studies reported that individuals with greater EUR possessed efficient SSC function [35, 37–42, 45, 58–62] and could be labelled as 'springy' [59], while individuals with lower EUR exhibited reduced SSC efficiency [37–42]. Given that EUR is a ratio- or percentage-based formula comparing an SSC and a non-SSC task, the added countermovement tends to consistently yield higher absolute values for any evaluated variable [57, 217], consequently increasing overall EUR values. However, higher EUR may not necessarily be beneficial as it could be associated with increased muscle slack and a reduced capacity to generate force rapidly [58]. Furthermore, improved performance in an SSC task can be attributed to the preceding countermovement allowing more work to be performed (i.e., impulse), rather than the result of being 'springy' through effective storage and reutilisation of elastic energy [217]. While a higher EUR may indicate efficient SSC function, it may also reflect factors that do not contribute to optimal performance, as previously discussed, warranting a nuanced interpretation of EUR values. Contrastingly, some research groups provided opposing claims indicating that lower EUR represents more efficient SSC function, while greater EUR indicates less efficiency [43, 44]. Across the studies, EUR was represented using common (i.e., force, height, power) [17, 33, 35–42, 45, 57–62] and alternative (i.e., impulse, net joint power/moment) [43, 44] performance variables. The varied use of these variables likely explained the contrasting findings for the mixed interpretations of EUR values.

Regardless, these alternative methods may still provide insight into one's movement strategy. For example, individuals exhibiting greater EUR (measured using net joint power and moment) was indicative of a movement strategy that was more 'hip- and/or ankle-dominant' [57, 59]. Based on these findings, it has been proposed that SSC utilisation in such cases may not rely purely on the elastic properties of the muscle–tendon unit (MTU), but rather on one's ability to actively generate force and produce joint work—particularly at the hip and ankle joints—potentially facilitated by spinal reflexes (i.e., muscle spindle stimulation) [57, 217, 218]. However, SSC function in jumping also depends on the ability to generate positive work at the knee joint, which has been associated with improved DJ performance [219]. These joint contributions are likely task- and strategy-dependent, and vary across individuals. Therefore, caution should be exercised when labelling individuals with greater EUR as having efficient SSC function or being 'springy', as such labels may oversimplify the complexity of underlying neuromuscular strategies.

In terms of stiffness, individuals who presented greater EUR [37, 39–41, 58] possessed lower extremity stiffness and were termed 'compliant' [37, 39–41], while lower EUR

counterparts were labelled as 'stiff' [33, 37, 39–41]. Stiffness across the studies was quantified through either calculation (i.e., vertical hop test) [33, 37, 39–41] or measured electronically (i.e., myometry) [40, 41]. However, such methods are limited based on existing assumptions in the methodology for limb stiffness calculation [41] and do not represent actual MTU stiffness [216]. Therefore, terms/labels of 'compliant' or 'stiff' lower extremity based on EUR values need to be interpreted with caution.

Individuals with greater EUR demonstrated enhanced force production, with JH used as a calculation variable and expressed as a percentage [37, 42]. A countermovement preceding a CMJ allowed the extensor muscles to build up active state and force prior to shortening, thereby increasing jump performance [217]. Given how EUR is calculated (i.e., CMJ variable as numerator and SJ variable as denominator), greater output values (i.e., greater CMJ performance) would then theoretically represent an individual's strength ability, supporting the findings above. Contrastingly, lower EUR was presented as a 'strength-dominant' profile, measured by using PF as a ratio [45]. Such differing findings could be a result of the varying measurement methods (i.e., percentage and ratio) and variables (i.e., JH and PF). Noteworthily, Kozinc et al. [36] established that greater EUR were indicative of 'less explosive' individuals and questioned the conventional interpretation of EUR as a straightforward marker for superior athletic performance. These inconsistencies might have arisen from measuring the EUR using performance variables other than those originally intended (i.e., PP instead of JH), potentially influencing the reported SSC efficiencies across studies. It is evident that EUR calculated from different methods and variables may not necessarily provide the same information about one's use of the SSC [35]. Hence, there is no good evidence to support labelling EUR values with respect to strength abilities.

EUR has inherent limitations that might affect its use in practice and, consequently, its interpretations. While the SSC concept is widely acknowledged and understood across sports science, directly quantifying SSC efficiency can be challenging and may involve some level of subjectivity. From a methodological standpoint, the body's centre of mass (COM) and joint moments often differ between CMJ and SJ during the push-off phase [217]. To enable valid comparisons, both movements should ideally begin from the same joint ROM (i.e., standardised hip, knee, and ankle angles); however, achieving this level of control is difficult in applied settings. Furthermore, it is known that differences between CMJ and SJ cannot be solely attributed to SSC utilisation [217]. This concern is compounded by the fact that EUR calculations typically rely only on performance outcomes (i.e., JH) and instantaneous variables (i.e., PF, PP), while disregarding any temporal aspects [35]. As such, one might question whether the EUR meaningfully reflects SSC utilisation.

The EUR purportedly evaluates one's SSC efficiency in athletic performance, but this assumption may not always hold true. While higher EUR values are generally associated with efficient SSC function (i.e., 'springy'), they do not entirely indicate superior performance due to potential underlying factors as previously mentioned. The conflicting findings regarding EUR's relationship with SSC efficiency and strength highlight the need for careful interpretation, particularly given the variations in measurement methods and performance variables used across studies. Furthermore, given the inherent conceptual and methodological limitations that the EUR method possesses, labelling individuals based on EUR values needs to be reconsidered.

5.1.1.3 Reactive Strength Index The RSI assesses one's ability to transition from an eccentric to a concentric muscular contraction reactively (during a DJ), and is commonly represented as a ratio of COM displacement or flight time to GCT (i.e., DJ height/GCT; flight time/GCT). Themes of *'adaptation'*, *'movement strategy'*, *'stiffness'*, and *'strength'* were identified across included studies.

Research indicates that greater RSI values were typically achieved due to greater neuromuscular adaptations [64, 69, 72, 73, 76, 79, 82, 83, 103, 113, 117, 121, 127], contributing to greater RFD [69, 107], or what some claimed to be 'explosiveness' [131] or 'explosive' abilities [100, 125, 126]. Some claims were based on logical inductions, where authors concluded that experimental interventions resulted in neuromuscular adaptations without directly measuring variables like RFD [69, 100, 107, 125, 126]. While RSI has been proposed as an index of 'explosiveness' due to its strong relationship with the rate of torque development in weightlifting performance [131], a clear definition of 'explosiveness' was not consistently provided. Given that these conclusions were largely drawn from indirect evidence, it is essential to reassess the classification of individuals with higher RSI as inherently more 'explosive'. The term 'explosive' refers to the ability to rapidly increase force or torque during the early phase of a voluntary contraction [220], commonly assessed within the first 200 ms following contraction onset [134]. The term has been applied to describe quickness, initial movement, expression of power, and other variations, making it more suited for coaching cues rather than as a movement classification [31]. Although some opinions suggest that the term 'explosive' should not be used to describe human movement [32], it remains a valuable and widely used descriptor that is unlikely to be dismissed. This distinction highlights the need for caution when using the term 'explosive' in research and underscores the importance of precise definitions and direct measurements, particularly when linking RSI to neuromuscular adaptations and 'explosive' capabilities. This may involve directly measuring RFD

and clearly defining what constitutes 'explosiveness' in the context of RSI.

Individuals with greater RSI values—typically defined relative to comparative groups within studies—were reported to exhibit efficient SSC utilisation [40, 71–75, 79, 80, 83, 84, 88, 89, 92–94, 97, 99, 106, 108, 109, 111, 114, 120, 122, 124, 128–130], and in some cases, were described as having 'good SSC function' or 'spring-like' qualities [106, 109, 128]. Contrastingly, lower RSI counterparts possessed inefficient SSC utilisation and were termed as having 'poor SSC function' [75, 79, 80, 92, 93, 97, 106, 109–111, 116]. Though most identified sources relied on logical inductions to justify their claims, there have been several attempts to quantify SSC function. Some authors identified individuals with the ability "to store and release mechanical energy" as having 'good spring function' through a novel joint function assessment [128], while other sources categorised participants as having 'good', 'moderate', or 'poor SSC function' based on the identification of impact peak and spring-like behaviour in DJ *F–T* curves [106, 109, 110]. Here, individuals categorised with 'good SSC function' were identified with landing and take-off PF occurring simultaneously [109], demonstrating their ability to seamlessly transit between eccentric, amortisation, and concentric movement phases. In this context, the term 'springy' aptly describes athletes who exhibit 'bouncy' and 'elastic' movement patterns, making them highly effective in movements that rely on rapid and efficient energy transfer. Measurable evidence here supports the notion that individuals with greater RSI, characterised by efficient SSC utilisation, can be labelled as 'springy', reflecting their superior ability to harness and transfer energy during dynamic movements (i.e., efficient force production relative to GCT). However, it is important to note that such terminology is metaphorical in nature. As McBride [218] highlights, maximal effort performance in hopping-based movement tasks is primarily driven by active muscle actuators rather than passive elastic mechanisms, underscoring the central role of muscular force generation in SSC-related movements. Therefore, while 'spring-like' profiles may serve as a useful descriptive term/label in applied settings, they may oversimplify the underlying neuromechanical strategies and should be interpreted with caution.

Being a ratio-based formula, RSI needs to be considered in conjunction with its constituent components that make up the value [70]. It is concerning that 33 out of 72 identified sources did not report constituent variables. Assessment of the respective variables can further provide insight to the strategy utilised to obtain a given RSI value. For instance, greater RSI could result from improved JH [70, 118, 119, 124], reduced GCT [74, 85, 87, 99, 103, 107, 120], or a combination of the two [68, 108]. Albeit in only two studies, greater RSI was also identified with reduced ROM [85, 102], which theoretically allows one to have a shorter GCT to then improve RSI. Specifically, GCT was kept below a threshold of 250 ms to measure reactive rather than concentric strength ability [66, 101, 105]. Though well established and commonly utilised, such a threshold should be further considered given that instructional cues, drop height, landing styles, and plyometric training history may all affect the resultant components and, ultimately, the RSI value [70, 85, 107, 120]. It should be questioned whether it is even necessary to differentiate 'fast SSC' and 'slow SSC' ability when the main objective is to assess inherent movement characteristics an individual may possess. In addition, JH reported were estimated from COM flight time [40, 71, 74, 87, 89, 91, 98, 102, 107, 109, 110, 114, 115, 118, 121, 122, 129, 132] or derived using the vertical velocity of the COM at take-off [84, 101, 103, 105, 108, 111, 116, 120, 124, 126]. Given that differences exist in JH obtained across varying methods [221], practitioners should be mindful of JH selection to ensure accurate and meaningful interpretations of the resulting metrics. Ultimately, it is essential to standardise assessment methodology to firmly ascertain that any variation in RSI reflects true differences in physical capabilities rather than being attributed to inconsistent methodologies.

Generally, individuals with greater RSI exhibited greater lower extremity stiffness and were inferred to be 'stiff' [40, 69, 72–75, 77, 79, 80, 82–84, 87, 88, 90, 92–94, 96, 102, 104, 108, 113, 114, 119, 123, 128–130], as opposed to individuals with lower RSI, who mostly possessed reduced limb stiffness or were termed 'compliant' [40, 84, 92, 93, 102, 104]. Sixteen studies measured lower extremity stiffness [40, 77, 80, 83, 84, 87, 88, 90, 92, 93, 104, 108, 114, 123, 129, 130], while the remaining 13 [69, 72–74, 76, 79, 82, 94, 96, 102, 113, 119, 128] provided logical induction to substantiate their claims. As noted in Sect. 5.1.1.2 (EUR), stiffness measurement has its inherent limitations [40, 216]. It is important to differentiate between methods of stiffness calculations, as they should not be used interchangeably. For example, in rebound tasks like the DJ used to assess RSI, ankle stiffness predominantly influences lower limb stiffness, particularly with shorter GCT [216]. Therefore, labelling individuals with greater RSI values as 'stiff' should be done with careful consideration, as these values are typically interpreted relative to other groups within a given study (e.g., intervention, strength level, or training status), and such comparisons may overlook joint-specific mechanics or movement strategy.

Another prominent finding across sources was that individuals with greater RSI possessed greater strength, demonstrating enhanced force production or attenuation abilities [6, 63, 65, 67, 68, 71, 76–80, 86, 90, 95–98, 112, 113, 115, 117–119, 124, 127, 132], and were termed 'strong' [63, 67, 68, 76, 78–80, 86, 95, 97, 112, 113, 118, 119]; lower RSI individuals typically exhibited reduced force production or attenuation abilities [63, 67, 68, 75, 78–81, 86, 96, 98, 110,

115], and were labelled 'weak' [63, 67, 68, 78–81, 86, 110]. Most studies presented some form of strength measure (e.g., 1 repetition maximum, braking/propulsion forces, isokinetic strength) to substantiate the assertion [6, 67, 68, 78–80, 86, 90, 98, 110, 115, 117–119, 124, 127, 132], while others provided no direct mechanistic evidence to support their claim [63, 65, 71, 75–77, 81, 95–97, 112, 113]. These findings align with Jarvis et al. [222], who identified that stronger individuals achieved higher RSI values, although the relationship was only moderate, suggesting that additional factors also contribute to RSI performance. While individuals with higher RSI may often exhibit greater strength abilities, it is not accurate to term/label them as 'strong' solely based on RSI values. It may be correct to say that they demonstrate effective rebound jump ability, which is often supported by strength, and underpinned by a combination of morphological and neural factors [223]. This illustrates the need for a more comprehensive approach to understanding RSI and its constituent components.

Taken together, findings indicate that individuals with greater RSI were frequently described using terms such as 'springy' or as exhibiting efficient/good SSC function in the included studies. However, these labels should be interpreted cautiously, as they may not accurately reflect specific neuromechanical mechanisms. Importantly, RSI alone does not imply traits such as 'explosiveness' (i.e., RFD), stiffness (i.e., 'stiff' or 'compliant'), or strength (i.e., 'stronger' or 'weaker'). When constituent components are factored into the analysis, movement strategies can be further elucidated (i.e., whether GCT was short or long). Overall, RSI should be examined with other NP methods or variables that more directly quantify specific physical qualities.

5.1.1.4 Reactive Strength Index Modified The RSImod assesses how efficiently one produces a momentum change, factoring in both the eccentric and the concentric movement phases during a CMJ. It is commonly represented as a ratio of JH or flight time to TTT (i.e., JH/TTT; flight time/TTT). Based on the findings, RSImod was associated with the recurring themes of *'adaptation'*, *'movement strategy'*, and *'strength'*, reflecting underlying factors such as efficient SSC function, greater RFD, rapid unloading, and enhanced force production, as interpreted across the included studies.

The results showed that individuals with greater RSImod exhibited efficient SSC utilisation [35, 44, 124, 133, 141, 145, 149, 154], as opposed to lower RSImod individuals who possessed inefficient SSC utilisation [44, 149, 154]. Identified studies only provided logical induction for the above assertion [35, 44, 124, 133, 141, 145, 149, 154], associating efficient SSC utilisation predominantly with movement strategies, which is further discussed in this section. For that matter, it may be premature to associate RSImod with SSC utilisation without assessing movement strategies involved, as these strategies may influence the constituent components that ultimately affect the RSImod.

Those with greater RSImod were observed to possess greater RFD [15, 134, 140, 143, 144, 153, 154, 162, 165], which aligns with them being labelled as 'explosive' [15, 126, 134, 143, 162, 163, 165], compared to lower RSImod counterparts who had reduced RFD production [154]. Studies supported this claim by measuring various RFD variables. Common RFD variables included early RFD (200 ms via IMTP) [134], peak RFD [143, 154], unloading RFD [140, 165], unloading and eccentric RFD [162], eccentric (or braking) RFD [144], as well as both eccentric and concentric RFD [140, 153]. Since RFD can be measured at different points corresponding to distinct phases of a movement on the *F–T* curve, no single RFD measure definitively represents 'explosiveness' in relation to RSImod values. Therefore, standardising RFD variables and the phases from which they are derived is crucial for more targeted and meaningful interpretation of an individual's abilities. Since 'explosiveness' describes the initial movement phase [31], it is suggested that RFD should be measured during the initial time epoch of a movement (i.e., RFD from 0 to 200 ms) [134]. However, it is important to understand that RSImod is not a direct proxy for RFD. Though RSImod has been described as a measure of 'explosive' strength [15, 143, 162, 163], and may be associated with RFD [134, 162, 163], it is important to regard both as distinct measures. With that said, RFD metrics—particularly during early contraction phases—often exhibit lower reliability [220], which should be considered when interpreting associations between RFD and RSImod in applied or comparative contexts. While these reliability concerns and the specific phases at which RFD is measured are important considerations, a detailed analysis lies beyond the scope of this review. Nevertheless, the assessment of RFD offers additional insight into an individual's ability to rapidly generate force (i.e., 'explosive' abilities). Therefore, an individual's 'explosiveness' can be partially inferred from the magnitude of their RSImod.

In terms of movement strategy, individuals with greater RSImod typically produced quicker countermovement (i.e., faster unloading) [23, 26, 111, 133, 137, 139, 140, 142–144, 155, 157, 158, 161, 165] and increased take-off velocity, which resulted in better jump performance [26, 111, 124, 139, 142, 147, 149–153, 155, 160, 161]. In contrast, individuals with lower RSImod achieved slower unloading [111, 137, 139, 142, 165] and reduced jump performance [23, 111, 139, 142, 147, 149, 153]. Across studies, RSImod was calculated using various definitions of TTT, often termed differently (e.g., contraction time, movement time), despite describing a similar duration from movement onset to take-off. These differences stem largely from variability in how movement onset and take-off thresholds were defined, rather than differences in the movement phases themselves.

Some sources examined only the unloading phase (initiation of countermovement to the lowest/minimum vGRF) [133, 139, 144]; others assessed both the unloading and braking phases (initiation of countermovement to zero velocity of the COM) [137, 140, 158]; and the remaining measured duration across a combination of the unloading, braking, and propulsive phases (initiation of countermovement to take-off; TTT) [23, 26, 111, 142, 143, 157, 161, 165]. Since unloading strategy directly influences take-off velocity and consequently, JH [133, 139], it is crucial to standardise the subphases within the CMJ, particularly in how duration is measured, before making any comparisons. The inconsistent definition of these movement phases presents challenges for practitioners, prompting recommendations that the phases of unloading, braking, and propulsion be adopted for more accurate CMJ assessment [224]. Considering the ratio-based nature of RSImod, an increase in JH would theoretically improve its overall value, assuming TTT reduces or remains unchanged. Additionally, if TTT decreases while JH remains the same or increases, RSImod can still increase, emphasising the need to consider both JH and TTT when interpreting this metric. Given how JH were obtained using different methods across sources (i.e., COM flight time and velocity at take-off) [15, 35, 101, 139, 140, 162], its calculation should also be standardised for appropriate comparison, as alluded to in Sect. 5.1.1.3 (RSI). Overall, the evidence here suggests that individuals with quicker countermovement unloading or increased jump performance can be represented through an increased RSImod. Deeper countermovement (i.e., greater ROM) has been associated with greater RSImod, as calculated via direct quantification of countermovement displacement via double integration of the *F–T* curve [26, 145, 148]. There was also contrasting evidence indicating shallower countermovement (i.e., lesser ROM) induced greater RSImod [150, 153, 155, 156]. Inconclusive findings may be attributed to the different instructions given to the participants to control the countermovement ROM [155, 156]. Therefore, it would be essential to standardise methodologies before such claims can be ascertained.

Some sources indicated that individuals with greater RSImod were 'stiff' [134, 141, 159], while others proposed otherwise (i.e., lower RSImod as 'stiff') [148]. However, claims here were justified qualitatively, without direct measurement of stiffness [134, 141, 159]. Although vertical stiffness was assessed in one study [148], it provided only an approximation of lower extremity stiffness, indirectly estimating COM displacement rather than MTU deformation [216]. Therefore, there is insufficient evidence to conclusively link RSImod with stiffness.

Unanimously, the results indicated that individuals with greater RSImod possessed enhanced force production capabilities [23, 48, 134–136, 138, 139, 141, 145–149, 153–155, 158–162] and were termed 'strong' [48, 134, 138, 141], whereas those with lower RSImod exhibited reduced force production capabilities [134–136, 138, 154] and were described as 'weak' [134, 138]. These studies substantiated the claim assessing force production capability through either kinetic jump variables (e.g., concentric mean force and/or PF) [23, 124, 139, 147, 148, 153–155, 160–162] or IMTP PF [134, 138, 145, 146]. Given that RSImod is calculated as the ratio of CMJ height to TTT, and that force production is one of the key determinants of JH [217], it is important to distinguish between reported values of peak and mean force. PF represents the maximal instantaneous value recorded, whereas mean force reflects the average force applied across the movement duration. CMJ height alone does not capture how force is generated, and its value can be influenced by factors such as countermovement depth and timing strategy [155]. This is further supported by Bobbert et al. [217], who demonstrated that CMJ performance exceeds that of SJ because the countermovement allows time for force buildup prior to muscle shortening, resulting in greater mean force application early in the concentric phase. This enhanced impulse (i.e., force applied over time), rather than PF alone, plays a more direct role in determining JH. While PF can be informative, it may not fully reflect an athlete's ability to apply force effectively if it occurs late in the movement, or is not sustained [217]. This reinforces a previously discussed limitation of the EUR, in that greater CMJ performance may reflect enhanced impulse development rather than superior eccentric utilisation (Sect. 5.1.1.2). Although individuals with greater RSImod are often classified as 'stronger' due to their enhanced force production, this interpretation should be made with caution. RSImod may reflect certain strength-related qualities, but does not fully capture the complexity of neuromuscular capacity—particularly when based solely on peak or mean force measures.

Across all evidence, the determination of an 'optimal' RSImod requires a comprehensive consideration of movement strategies such as unloading speed, countermovement depth, and force production abilities, all of which contribute to the intricate interplay affecting JH (influenced by take-off velocity and force production abilities) and TTT (influenced by unloading speed and ROM). Consensus points to greater RSImod being 'explosive', 'strong', and producing a deep countermovement with quicker unloading. In theory, the greater the COM traverses during a CMJ (vertical displacement), the longer the duration will be to take-off, effectively reducing the RSImod. It is possible that 'strong' and 'explosive' individuals can produce deep countermovement in a short enough duration (i.e., quick unloading) to induce greater RSImod [26], supporting the use of these terms/labels. Effectively, RSImod should still be considered alongside other NP methods or performance metrics that offer insight into both the temporal and force production aspects of athletic performance.

5.1.2 Non-Ratio-Based Neuromuscular Profiling Methods

5.1.2.1 Force–Velocity Profile Force–velocity profiling is a concept that examines the relationship of force and velocity values obtained through field-based measures of either a sequentially loaded vertical jump test [18] or a horizontal sprint task [196]. It is based on the notion that force and velocity are inversely related in human motion [225], and that theoretical optimal profiles exist and can be determined to maximise performance within individuals [18, 196]. Based on the concept, maximum force (*F*0) and maximum velocity (*V*0), which indicate what an individual can theoretically express, are identified and interpreted as their FVP through the gradient of the linear force–velocity relationship (known as S*fv* in the literature) and are represented by the formula $-F0/V0$ [21]. An individual's optimal FVP (indicated as Sfv_{opt}), representing the ideal balance between force and velocity capabilities allowing for maximal power production ($P_{max} = F0 \times V0/4$), can further be determined [21, 181]. From here, the relative difference between the actual and optimal FVP for an individual can be calculated as their force–velocity imbalance (i.e., $FV\text{imb} = Sfv/Sfv_{opt} \times 100$), with 100% representing a well-balanced profile [21, 181]. Profiles that exhibit a *FV*imb greater or lower than the optimal FVP (i.e., 100%) would result in a performance deficit in velocity and force capabilities, respectively [183].

Across the included studies, themes of *'movement strategy'* and *'strength'* came through as prominent. This is no surprise given that in a linear force–velocity relationship, a steeper slope (i.e., more negative S*fv* values) would indicate that a profile has greater force production ability given an increased F0, while a gentler slope (i.e., less negative S*fv* values) would highlight a profile that is more 'velocity-oriented' given an elevated V0. This is consistent across most studies that examined the FVP, labelling individuals with lower S*fv* (i.e., steeper slope or more negative values) as 'force-oriented' or 'force-dominant' [18, 21, 168, 169, 173–176, 178, 180–190, 192, 195–198, 200, 202–204], and those with greater S*fv* (i.e., gentler slope or less negative values) as 'velocity-oriented' or 'velocity-dominant' [18, 21, 56, 166, 170, 174–178, 181, 183, 185, 186, 193–196, 198, 200–202]. One exception identified was that profiles exhibiting a gentler slope were misclassified as 'force-oriented' when they should be considered otherwise, given that S*fv* values were more negative in profiles with a gentler slope (S*fv* = − 0.80 to − 1.15 Ns/m/kg) compared to those with a steeper slope (S*fv* = − 0.75 to − 1.10 Ns/m/kg) [178]. Misclassification could stem from mistaking and associating a higher S*fv* value (a profile of gentler slope) with one that is considered as lower, albeit it being more negative numerically, when in fact it should be the opposite.

'Force-oriented' individuals were also termed 'terrestrial' [199], while velocity-oriented individuals were labelled 'aerial' [199], which some have inferred as 'explosive' [179, 189]. Authors [199] grouped participants using a validated field method (Volodalen method) into 'aerial' and 'terrestrial' runners, compared that with subsequent horizontal FVP assessed, and provided further insight between the two groups. 'Terrestrial' runners observed more prominent rearfoot striking patterns, contributing to greater horizontal propulsion, making them more adapted to specific short distance and high acceleration sprints, whilst 'aerial' runners were more effective in maintaining a net horizontal force production with increasing speed and were more suited to longer sprints [199]. The results obtained were congruent with horizontal FVP findings from Buchheit et al. [171] and Morris et al. [191], with the 'acceleration-focused' group being associated with greater force capability, and the 'maximal sprinting speed-focused' group being associated with greater velocity and/or power capabilities. In the context of horizontal FVP, 'velocity-dominant/oriented' individuals may logically be associated with 'explosiveness' due to their higher power and speed capabilities. Furthermore, findings here consistently indicated that profiles with *FV*imb above 100% of the optimal FVP reflect a 'velocity-deficit' (i.e., reduced velocity capability), while profiles with *FV*imb below 100% of the optimal FVP exhibit a 'force-deficit' (i.e., reduced force capability).

Recent studies have raised concerns about the use of FVP and the validity of its underlying concept [226–228]. One key criticism is that FVP may oversimplify the complex relationship between force and velocity, leading to potential misinterpretations of an athlete's physical capabilities. For example, FVP relies on linear extrapolations to estimate one's 'optimal' profile, but these extrapolations may not account for variability in movement patterns, muscle activation, or joint mechanics [226–228]. Furthermore, these extrapolated variables can be compromised by even slight experimental variations, making them unreliable if testing conditions are not appropriately standardised [229]. In addition, categorisation of athletes into 'force-dominant' or 'velocity-dominant' based on FVP does not always reflect their true capabilities, as these imbalances may stem from other factors, such as coordination or neuromuscular efficiency, rather than from an inherent deficiency in force or velocity production [226–228]. Rather than relying on these theoretical extrapolations, a more practical approach could involve calculating force at zero velocity directly during an isometric test. This method would provide a more accurate and reliable measurement of strength abilities [212, 213], as it avoids the pitfalls of extrapolation and may be less susceptible to variability in testing conditions. Consequently, future research could benefit from comparing these predicted maximal force values with those obtained through direct isometric testing, which may provide a more accurate

assessment of an athlete's strength capabilities. While the FVP offers means of distinguishing profiles among individuals, it is important to note that its conceptual association may be limited to the specific tasks assessed (i.e., vertical jump, horizontal sprint), and may not necessarily reflect an individual's inherent maximal force or velocity capacities [226, 227]. Despite the limitations, FVP may still provide some insight that could inform training interventions [181], albeit interpretation should be approached with caution.

Understanding one's FVP can better facilitate the optimal prescription of resistance training, enabling targeted improvements in either force or velocity aspects across individuals [166, 175, 181, 183]. Consensus showed that a steeper *FV*slope/*Sfv* (i.e., more negative values) or *FV*imb > 100% of the optimal FVP represent profiles that are 'force-dominant/oriented' or exhibit 'velocity-deficit' due to greater force capability. In contrast, profiles with gentler *FV*slope/*Sfv* (i.e., less negative values) and *FV*imb < 100% of the optimal FVP were regarded as 'velocity-dominant/oriented' or as portraying 'force-deficit' because of greater velocity capability. Although FVP offers insight into one's abilities, it encompasses limitations surrounding its applicability and validity [226–228], and should be interpreted with caution when utilised as a profiling method.

5.1.2.2 Impulse Analysis With the growing popularity in force platform usage, researchers have increasingly employed the concept of IA, focusing on the evaluation of components derived from *F–T* curves typically acquired during dynamic tasks (i.e., CMJ and DJ). *F–T* curves are generally broken into phases to allow for subsequent analysis through statistical means such as cluster analysis, factor analysis, principal component analysis, or temporal phase analysis [20, 108, 143, 154, 155, 205, 207]. Though complex and unfamiliar to many practitioners, these methods contribute to allow an enhanced understanding of the evaluated task by simplifying complex data and revealing patterns along with their underlying relationships [20, 44, 143, 206]. Given that these methods examine patterns and relationships across diverse dimensions within the data, findings here span across a broad spectrum of themes. Although impulse is not calculated directly as a discrete variable in all cases, the analysis presented here is grounded in the interpretation of the *F–T* curve, through which impulse-related characteristics (i.e., force and temporal variables) were evaluated.

Several studies grouped participants as having 'bimodal' or 'unimodal' patterns identified based on the shape of the *F–T* curve [20, 44, 154, 155, 164, 207, 209, 211]. Briefly, 'bimodal' provides an indication of two distinct and separate force peaks while 'unimodal' refers to only one distinct force peak during the propulsion phase of the CMJ [155]. While some authors regarded individuals with a 'bimodal' pattern as having efficient SSC function [20, 154], there were others that suggested otherwise [44, 211]. Currently, there is no consensus regarding the underlying mechanisms behind the presence of either a 'bimodal' or 'unimodal' force pattern given that human movements are complex and can be influenced by varying factors. Therefore, profiles based on these two terms/labels do not offer relevant insight to support additional training.

Individuals who achieved optimal jump performance or greater RSI and RSImod values often exhibited efficient SSC utilisation [20, 44, 106, 108–110, 154, 205, 208, 211] or 'spring-like' profiles [106, 109, 110]. These individuals also demonstrated greater RFD measured across different phases of the jump [20, 143, 154] and were often described as 'explosive' [143, 163]. As alluded to in Sect. 5.1.1.4 (RSImod), the identification and standardisation of RFD variables could provide targeted interpretation of specific performance traits. However, studies that evaluated RFD often inferred SSC function based on qualitative inferences rather than direct measurement [20, 143, 154]. Although elevated RFD or 'explosiveness' is often logically associated with 'efficient SSC function' or 'spring-like' profiles, caution is warranted in interpreting these links, as they may oversimplify complex neuromechanical processes, as discussed in Sect. 5.1.1.3 (RSI). Overall, the findings here aligned with observations outlined in Sects. 5.1.1.3 (RSI) and 5.1.1.4 (RSImod), where greater RSI and RSImod values were frequently associated with descriptors such as 'explosiveness' and 'efficient SSC function' in the reviewed literature. However, these associations should be interpreted cautiously, as they are largely based on inferred reasoning rather than direct mechanistic evidence.

It is understood that individuals with deeper and faster countermovement tend to exhibit 'efficient SSC function', thereby enhancing force production and attenuation capabilities, contributing to greater jump performances [205]. This efficiency can be attributed to the work done and contribution through spinal reflexes (i.e., muscle spindle stimulation) [230]. Contrastingly, there were others that associated shallower ROM with good SSC function [110]. Theoretically, the variation in either countermovement depth and/or speed affects the time available to apply force, altering the impulse, which directly contributes to CMJ performance [222]. Consequently, a combination of a deep and fast countermovement facilitates enhanced force production over increased ROM, and hence impulse, ultimately enhancing jump performance [143, 164, 205, 206, 208]. When PF is identified early in the jump phases, particularly coinciding with a low COM position (i.e., end of the braking phase) [164, 209], this results in greater braking impulse [23], which may then contribute to greater propulsive impulse, take-off velocity, and thus JH [147].

Overall, the findings point to the importance of the interplay between *F–T* curve patterns, movement characteristics, and optimal temporal force application across jump analysis. Regardless of whether the *F–T* curve exhibits a 'bimodal' or 'unimodal' pattern, the key to an efficient CMJ lies in executing a deep and fast countermovement facilitating early PF and impulse production [164, 209]. This further contributes to greater accelerative and propulsion abilities, increasing take-off velocity, and consequently JH [20, 205]. In the context of practicality, when there is already an abundance of information available for assessment, the myriad of variables accessible by the IA method—such as *F–T* curve patterns, movement characteristics, and temporal force application—could potentially overwhelm practitioners, particularly when the profiling objective is to be both comprehensive and time-efficient. Given the practical considerations and complexities associated with implementing IA, it should be applied selectively, depending on the specific objectives of athlete assessment.

5.2 Concluding Statements Concerning Neuromuscular Profiling Methods

This review represents the first systematic effort to critically examine how output values from empirically based NP methods are grouped, labelled, and interpreted. While each method offers a specific testing focus or perspective on neuromuscular characteristics, findings revealed considerable variability in terminology, calculation methods, and interpretive frameworks across studies. This overlap highlights the need for clearer reporting standards and standardised terminology to ensure practitioners can meaningfully compare outputs across methods, rather than interpreting similar traits under different labels.

In response to these challenges, and to support clearer interpretation and more standardised application, two summary tables were established based on the review's findings. Tables 4 and 5 were developed to enhance the accuracy and consistency of NP interpretation by consolidating commonly used terminology, conceptual definitions, methodological limitations, and associated interpretations, based on findings synthesised from Supplementary Tables 1.1–1.6 (ESM). Table 4 is a compilation of recurring terms and labels used across NP studies, and aligns them with their intended definitions and practical usage to reduce ambiguity. Table 5 presents a structured overview of each NP method, detailing its operational definition, measurement approach, key limitations, and associated terminology and interpretation. Together, these tables facilitate clearer cross-method comparisons, improve interpretability, and serve as a practical reference for both practitioners and researchers seeking to apply NP assessments more effectively in practical and research contexts.

5.2.1 Summary of Ratio-Based Neuromuscular Profiling Methods

Ratio-based NP methods, which include DSI, EUR, RSI, and RSImod, are commonly used to provide insight into the neuromuscular mechanisms underlying performance outputs. While these methods were frequently examined across studies, relatively few provided consistent terms/labels to describe the characteristics associated with different output values. Common terms/labels elucidated were: 'explosive/explosiveness', 'stiff/compliant', 'spring/springy', and 'strong/weak'. A summary of each NP method follows.

The DSI offers insight into strength qualities, but requires careful interpretation of both dynamic and isometric PF. Since DSI is based on instantaneous PF values, it overlooks temporal characteristics like impulse, which may better reflect dynamic performance. Thus, DSI should be interpreted alongside its constituent components and complemented with other NP methods—such as RSImod or IA—for a more complete neuromuscular assessment.

EUR aims to evaluate one's ability to utilise the SSC, but there are inherent limitations that might affect its application, and consequently, its interpretations. Practical constraints—such as the need to standardise jump assessments—and the reliance on performance outcomes and instantaneous variables, while overlooking temporal components, raise questions about whether EUR accurately measures SSC utilisation. Therefore, utilisation of EUR should be carefully evaluated within the context of these limitations.

RSI and RSImod may serve as partial indicators of SSC function, RFD, or strength abilities, though these interpretations warrant caution. Individuals with greater RSI or RSImod values are often described as 'springy' (efficient SSC), 'explosive' (greater RFD), or 'stronger' (enhanced force production). However, these labels are frequently based on indirect inference rather than direct mechanistic evidence. Given their ratio-based nature, interpretation should always consider constituent components (e.g., JH, GCT, TTT), as similar output values may arise from different movement strategies. Specifically, RSImod is influenced by countermovement characteristics (e.g., depth, unloading speed), while RSI is more sensitive to contact time thresholds and landing technique. Additionally, RFD measures—especially when associated with RSImod—may exhibit limited reliability. Standardising assessment methodologies (e.g., jump instructions, JH and GCT/TTT calculations) is essential to ensure that output differences reflect true physical capabilities rather than methodological inconsistencies. Incorporating variables representing movement strategies (i.e., GCT and TTT) enhances the interpretability and utility of NP metrics.

Table 4 Definitions and scientific measurements of common neuromuscular profiling terms

Term/label	Definition	Scientific measurements
'Explosive' or 'Explosiveness'	Describes an individual's ability to generate force quickly, typically associated with rapid movements.	**Rate of force development (RFD):** Measured via force plates or load cells, ideally during initial phases of movement (e.g., 0–200 ms).
'Spring', 'Springy', or 'Efficient SSC Function'	Reflects the ability to utilise the elastic potential of the musculotendon unit effectively during dynamic movements.	**Reactive Strength Index/Reactive Strength Index-Modified (RSI/RSImod):** Measured using force plates, photocell devices, or contact mats. Higher RSI/RSImod ratios typically indicate one's ability to exhibit a quick and forceful dynamic movement (i.e., DJ, CMJ), minimising GCT or TTT while maximising vertical jump height.
'Stiff' or 'Compliant'	Describes the ability of the body to resist deformation (stiff) or to absorb and store energy (compliant).	**Stiffness measures:** Quantified through stiffness calculation during hopping or jumping tasks, often measured via force plates, capturing vertical stiffness or leg stiffness.
'Strong' or 'Weak'	Refers to an individual's maximal force production capability.	**Strength testing:** One-repetition maximum testing. **Peak force production:** Measures maximal force output, using force plates or dynamometers. **Force–Velocity Profile (FVP):** Can also indicate strength qualities.
'Force-Dominant/Oriented' or 'Velocity-Dominant/Oriented'	Reflects the balance between force and velocity during measured dynamic movements.	**Force–velocity slope (*FV*slope):** Steeper (more negative) slope indicates force dominance, gentler slope indicates velocity dominance, calculated through a load–velocity curve from FVP analysis. **Force–velocity imbalance (*FV*imb):** Imbalance relative to optimal FVP.
'Unimodal' or 'Bimodal'	Indicates the presence of one or two distinct force peak(s) during a dynamic movement, usually a CMJ.	**Force–time curve:** Analysed through force plates during dynamic movements like jumping or sprinting. Unimodal curves show a single force peak, while bimodal curves have two distinct peaks.
Joint Dominance (i.e., 'hip-dominant', 'knee-dominant', 'ankle-dominant')	Describes the relative contribution of specific joints (e.g., hip, knee, or ankle) to movement.	**Joint kinematics and kinetics:** Assessed using 3D motion capture systems and force plates, identifying joint angles, moments, and power contributions from specific joints during movement tasks.

The terms and associations listed in this table are derived from terminology trends observed across included studies and supplementary synthesis tables. They are intended to reflect common interpretive usage rather than serve as prescriptive or empirically validated definitions. These terms/labels are interpretive in nature and should be applied cautiously, particularly in light of the methodological limitations discussed throughout the review

CMJ countermovement jump, *DJ* drop jump, *GCT* ground contact time, *TTT* time to take-off

Table 5 Overview of neuromuscular profiling methods and interpretive considerations

Neuromuscular profiling methods	Concept and definition	Measurement approach	Key limitations	Terminology and interpretation
Dynamic Strength Index (DSI)	Provides insight into the relative expression of peak dynamic to isometric force production.	■ Typically calculated as the ratio of PF from a ballistic task (e.g., CMJ) to that from an isometric task (e.g., IMTP, ISqt). ■ Tasks should involve similar joint mechanics (e.g., both using triple extension) for meaningful interpretation. ■ Some studies have used impulse-based variants, but these are less common and not well validated.	■ Captures only instantaneous (peak) values, overlooking temporal aspects of force production (e.g., impulse, RFD). ■ DSI values vary based on the dynamic and isometric tasks selected. ■ Provides limited insight without complementary metrics (e.g., temporal data).	■ Interpretation should consider the underlying component values (e.g., peak dynamic and isometric force). ■ Terms like 'strong' and 'weak' are often applied based on DSI values but may misrepresent actual strength levels. ■ Labels such as 'explosive' or 'stiff' are occasionally applied but lack direct support from DSI's underlying components.
Eccentric Utilisation Ratio (EUR)	Intended to assess how effectively the countermovement contributes to performance, often used as an indirect indicator of SSC function.	■ Most expressed as a ratio, percentage, or difference between SSC and non-SSC task outcomes (e.g., JH). ■ While power and impulse have also been used, these are less common and not well validated.	■ Omits temporal variables, limiting its accuracy in reflecting actual SSC utilisation. ■ Valid comparisons require standardised starting ROM for CMJ and SJ, but this is difficult to achieve in applied settings. ■ JH is ideally calculated using Vto rather than FT. ■ Lack of standardisation in calculation methods and variable selection limits comparability across studies.	■ Higher EUR is often interpreted as indicative of better SSC efficiency or elastic utilisation, but such assumptions are inconsistently supported and may be confounded by the chosen measurement approach. ■ Labels such as 'springy', 'stiff', or 'compliant' are commonly applied but may oversimplify the underlying neuromuscular characteristics, which are typically inferred rather than directly measured. ■ Higher EUR may reflect greater strength capacities or joint-dominant movement strategies (e.g., hip- or ankle-dominance), but such interpretations remain unclear and inconsistently supported.
Reactive Strength Index (RSI)	Assesses the ability to transition from eccentric to concentric muscular contraction during a DJ task (i.e., reactive capability).	■ Calculated as FT or DJ height divided by GCT during a DJ. ■ Other variation includes rebound JH divided by GCT. ■ A GCT threshold (commonly < 250 ms) is often applied to target reactive characteristics.	■ Used to infer "fast SSC function", but this classification lacks direct mechanistic validation. ■ JH is ideally calculated using Vto rather than FT. ■ Interpretation requires analysis of constituent components (i.e., JH, FT, GCT).	■ Higher RSI is frequently interpreted as indicating superior SSC function or reactive ability, but such interpretations should be made cautiously and supported by analysis of whether performance was driven by shorter GCT, higher JH, or both. ■ Labels such as 'explosive', 'springy', 'stiff', or 'strong' are commonly used but are often based on indirect inference rather than direct measurement.
Reactive Strength Index Modified (RSImod)	Assesses efficiency of momentum change during CMJ by incorporating both eccentric and concentric phases.	■ Calculated as JH divided by TTT during a CMJ. ■ Other variations include FT divided by TTT and horizontal jump distance divided by TTT.	■ CMJ subphases (e.g., unloading) are inconsistently defined across studies and require standardisation. ■ Influenced by countermovement depth, unloading speed, and take-off strategy. ■ JH is ideally calculated using Vto rather than FT. ■ Interpretation requires analysis of constituent components (i.e., JH, TTT).	■ Interpretation should account for how movement strategies (e.g., unloading speed, countermovement depth) influence the resulting value. ■ Labels such as 'explosive' or 'strong' are commonly applied to individuals with higher RSImod values due to associations with improved jump performance and RFD; however, these traits are not directly represented by the constituent components and should be inferred with caution.

Table 5 (continued)

Neuromuscular profiling methods	Concept and definition	Measurement approach	Key limitations	Terminology and interpretation
Force–Velocity Profile (FVP)	Models the linear relationship between theoretical force and velocity outputs derived from ballistic tasks, typically to identify an individual's balance between force- and velocity-producing capabilities.	■ Calculated by deriving the *FV*slope ($Sfv = -F0/V0$) and comparing it to an optimal slope (Sfv_{opt}) to quantify force–velocity imbalance (*FV*imb). ■ Applied to ballistic tasks (e.g., jumps, sprints) using regression of task-specific force and velocity data.	■ Assumes a linear force–velocity relationship, with estimates (e.g., F0, V0) derived via extrapolation that can be compromised by the slightest experimental variations. ■ Extrapolated outputs are rarely validated against direct measures (e.g., isometric testing), and resulting categorisations (e.g., force- or velocity-deficient) may not reflect actual performance limitations.	■ Labels such as 'force-oriented', 'velocity-dominant', 'explosive', 'terrestrial', or 'aerial' are commonly applied based on *Sfv* and *FV*imb values; however, these categories are task-specific and often derived from extrapolated estimates rather than direct mechanical measurements. ■ Interpretation should be contextualised to the assessed task (e.g., jump vs. sprint) and ideally supported by complementary measures (e.g., strength testing, IA) to validate force–velocity characteristics.
Impulse Analysis (IA)	Evaluates force application over time during dynamic tasks by segmenting the *F–T* curve into discrete movement phases (e.g., eccentric, concentric) and analysing phase-specific metrics.	■ Measured and analysed through methods such as cluster, factor, or temporal phase analysis to identify performance-relevant patterns. ■ Typically applied across dynamic tasks (i.e., CMJ, DJ).	■ Method is complex to administer and interpret consistently. ■ Large number of available variables may overwhelm practitioners and hinder practical application. ■ Extracted information are often used to infer movement strategy but lack standardisation.	■ Curve-based classifications (e.g., 'unimodal', 'bimodal') are inconsistently defined across studies and lack mechanistic consensus. ■ Labels such as 'spring-like', 'explosive', or 'efficient SSC function' are commonly inferred from *F–T* curve patterns; however, these traits are not directly measured and should be interpreted with caution. ■ IA profiling should be applied selectively based on assessment objectives, considering *F–T* curve patterns, movement characteristics, and other NP metrics.

CMJ countermovement jump, *DJ* drop jump, *DSI* Dynamic Strength Index, *EUR* Eccentric Utilisation Ratio, *F–T* force–time, *F0* theoretical maximal force, *FT* flight time, *FVimb* force–velocity imbalance, *FVP* Force–Velocity Profile, *GCT* ground contact time, *IA* impulse analysis, *IMTP* isometric mid-thigh pull, *ISqt* isometric squat, *JH* jump height, *NP* neuromuscular profiling, *PF* peak force, *RFD* rate of force development, *ROM* range of motion, *RSI* Reactive Strength Index, *RSImod* Reactive Strength Index Modified, *Sfv/FVslope* force–velocity slope or profile, Sfv_{opt} optimal force–velocity slope or profile, *SJ* squat jump, *SSC* stretch–shortening cycle, *TTT* time to take-off (also known as contraction time), *V0* theoretical maximal velocity, *Vto* velocity at take-off

5.2.2 Summary of Non-Ratio-Based Neuromuscular Profiling Methods

Non-ratio-based NP methods, such as FVP and IA, can be regarded as independent conceptual frameworks that do not require integration with other NP methods to effectively describe an individual's profile. Terms/labels that were elucidated from non-ratio-based NP methods were: 'force-dominant/oriented or velocity-dominant/oriented'—indicating greater force or velocity capability; 'force-deficit or velocity-deficit'—representing reduced force or velocity capability; 'unimodal or bimodal'—providing information on the presence of one or two distinct force peak(s) on the *F–T* curve; 'hip-dominant/flexors or knee- and ankle-dominant/hyper flexors'—demonstrating joint dominance for a given movement. Respective findings are summarised as follows.

Consensus from FVP studies was that a steeper *FV*slope/*Sfv* (i.e., more negative values) or *FV*imb > 100% of the optimal FVP represent profiles that are 'force-dominant/oriented' or exhibit 'velocity-deficit' due to greater force capability. In contrast, profiles with gentler *FV*slope/*Sfv* (i.e., less negative values) and *FV*imb < 100% of the optimal FVP were regarded as 'velocity-dominant/oriented' or portraying 'force-deficit' because of greater velocity capability. However, the FVP concept is only limited to the specific tasks assessed and may not necessarily reflect an individual's inherent maximal capacities as is commonly claimed. Given the limitations surrounding the applicability and validity of FVP, insight should be interpreted with caution, particularly as the method requires multiple load conditions and standardised execution, which may present practical challenges in applied settings.

Even though 'bimodal' and 'unimodal' terms/labels appear as consistent across IA studies, there is no consensus regarding the underlying mechanisms behind the presence of either term/label given that human movements are complex and can be influenced by varying factors. Overall, the findings point to the importance of the interplay between *F–T* curve patterns, movement characteristics, and optimal temporal force application across identified IA studies. While practical challenges and complexities may limit its widespread use, IA offers a conceptually grounded approach that directly relates force application to performance outcomes, and may be underutilised in both research and practice.

5.3 Limitations

This systematic review has inherent limitations. First, this review involves inherent subjectivity in both the interpretation of article content and the synthesis of terminology. While some studies were explicit in how they classified neuromuscular characteristics, many were more implicit—requiring interpretive judgement from reviewers. Importantly, the included terms/labels (e.g., 'explosive', 'stiff', 'spring-like') were not selected a priori, but rather emerged through inductive coding of terminology as used in the included literature. Although this introduces interpretive bias, it reflects how NP outputs are described in current research and applied practice. To mitigate misinterpretation, we employed a rigorous process involving evaluation by at least two reviewers, with discrepancies resolved through discussion and consensus. While the synthesis remains interpretive, it provides a necessary foundation for consolidating NP terminology and critically examining how such language aligns—or fails to align—with the neuromuscular characteristics these methods are intended to represent.

Second, while the appropriate selection of NP methodology is important, it is also necessary to consider the reliability of NP methods along with the corresponding assessments to determine its constituent components. Out of the 184 included sources, only 47 (26%) reported some form of reliability statistics. The most reported statistics were the intraclass correlation coefficient (ICC), coefficient of variation percentage (CV%), and typical error (TE), interpreting them based on 95% confidence interval levels. Greater ICC values, lower CV%, and TE typically indicate greater consistency and precision in the assessment process. Factoring in reliability results will ensure that any change in NP method value is a true representation of an individual's true physical capabilities rather than methodological inconsistencies [8], further justifying empirically based NP methods as valuable tools in the profiling of athletes.

In addition to these study-level reliability concerns, it is also important to acknowledge limitations inherent to NP calculation approaches. Specifically, ratio-based metrics (e.g., DSI, EUR, RSI, RSImod) are susceptible to accumulated variability, as fluctuations in both the numerator and denominator can amplify measurement noise in the final output. This issue is particularly relevant for FVP, which requires multiple testing conditions (e.g., varying loads), thereby increasing the likelihood of compounding measurement error across trials. While IA was less frequently critiqued in this regard, it presents practical implementation challenges, including greater data processing demands and the potential to overwhelm practitioners due to the volume of output variables. However, if automated force plate systems are used, data processing becomes less of a barrier—practitioners can readily extract variables such as net impulse, mean net force, and phase duration to gain relevant insights. Instead, it is often the communication and contextualisation of these outputs with athletes and technical coaches that introduces additional time demands. Overall, the findings suggest that method selection should reflect the performance characteristics of interest, while balancing conceptual rigour, reliability, and practical feasibility.

6 Conclusion and Future Directions

This review represents a foundational step toward improving standardisation in NP interpretation by consolidating the terminology currently used across methods and examining its conceptual alignment with measured outputs. In doing so, it highlights the importance of developing clearer frameworks to enhance communication and practical application across both research and practitioner contexts. Moving forward, future research should examine whether the labels assigned to NP metrics accurately reflect the neuromuscular characteristics they intend to represent, particularly in relation to the subjective language commonly used by coaches and practitioners. Such investigations may help bridge the gap between empirical measurement and applied athlete profiling, ultimately enhancing the utility of NP in performance evaluation and training decision-making. Importantly, given the interpretive overlap observed across many NP methods, practitioners should not feel compelled to apply all methods concurrently. Instead, method selection should be based on the specific neuromuscular characteristics of interest, the testing context, and available time and resources. This approach balances conceptual relevance with practical feasibility, helping reduce redundancy while maintaining the value of NP in athlete monitoring. Additionally, while most existing NP methods focus on lower body tasks, future work should also aim to develop and validate NP approaches for upper body assessments. Doing so would allow for a more comprehensive evaluation of an athlete's overall neuromuscular function and broaden the applicability of profiling practices across different sporting demands.

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